

Types of Rectifiers:

1. Half wave Rectifier with R load.
2. Half wave Rectifier with R-L Load.
3. Half wave Rectifier with R-L Load and free Wheeling Diode.
4. Half wave Rectifier with RLE Load
5. 1- ϕ Full wave Mid Point Rectifier.
6. 1- ϕ Full wave Bridge type Rectifier with R-L-E Load.
7. 3- ϕ controlled Half wave Rectifier.
8. 3- ϕ controlled Full wave Rectifier.
9. 1- ϕ Semi converter with R-L-E Load.
10. 3 - ϕ Full wave Semi converter with RLE Load & Freewheeling Diode.
11. Effect of source Inductance on the performance of single phase Rectifier.
12. Dual converter

Introduction:

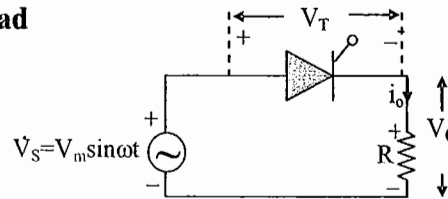
- In the study of thyristor systems, SCR's and diodes are assumed ideal switches which means that
 - (i) During turn on there is no voltage drop across them,
 - (ii) No reverse current exists under reverse blocking mode.
 - (iii) transition time = 0
 - (iv) Holding current is zero.
- Trigger circuits are not shown in SCR circuit for convenience.
- **FIRING ANGLE:** The firing angle is defined as the angle between the instant thyristor would conduct if it was a diode and the instant it is triggered.

(or)
- A firing angle is defined as the angle measured from the instant that gives the largest average output voltage to the instant it is triggered.

(or)
- A firing angle may be defined as the angle measured from the instant SCR gets forward biased to the instant it is triggered.
- By varying the firing angle α , the phase relationship between the start of the load current and the supply voltage can be controlled, hence the term phase control is

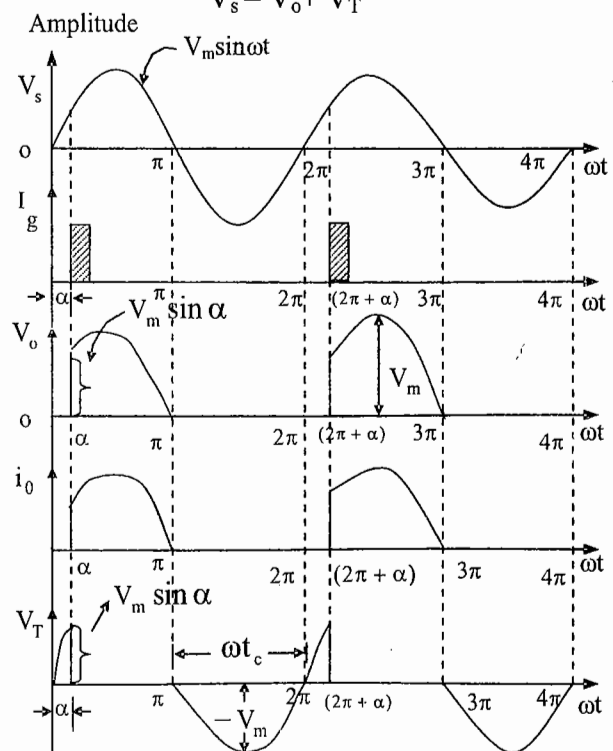
used for such a method of controlling the load currents.

1- ϕ Half Wave phase controlled rectifier with R - Load



- During the positive half cycle of the supply, SCR is forward biased, at firing angle ' α ' it starts conducting.
- Thyristor conducts from $\omega t = \alpha$ to π , $(2\pi + \alpha)$ to 3π and so on. Over the firing angle delay α load voltage $V_o = 0$ but during conduction angle $(\alpha$ to $\pi)$, $V_o = V_s$.
- As firing angles is increased from zero to π , the average load voltage decrease from the largest value to zero.
- The thyristor remains ON from $\omega t = \alpha$ to π , $(2\pi + \alpha$ to 3π etc.,) during these intervals $V_T = 0$. It is off from π to $(2\pi + \alpha)$, 3π to $(4\pi + \alpha)$ etc., during these off intervals V_T has the wave shape of supply voltage V_s .

$$V_s = V_o + V_T$$





- The SCR is reverse biased for π radians, the circuit turn off time is given by

$$t_c = \frac{\pi}{\omega}$$

Where, $\omega = 2\pi f$ and f is the supply frequency in Hz.

- Average voltage V_o across load R in terms of firing angle ' α ' is given by

$$V_o = \frac{1}{2\pi} \int_{\alpha}^{\pi} V_m \sin \omega t \cdot d(\omega t)$$

$$V_o = \frac{V_m}{2\pi} (1 + \cos \alpha)$$

- The maximum value of average output voltage V_o occurs at $\alpha = 0^\circ$.

$$V_{o\max} = \frac{V_m}{2\pi} (2) = \frac{V_m}{\pi}$$

- Average load current,

$$I_o = \frac{V_o}{R} = \frac{V_m}{2\pi R} (1 + \cos \alpha)$$

- RMS value of load voltage

$$V_{or} = \sqrt{\left[\frac{1}{2\pi} \int_{\alpha}^{\pi} V_m^2 \sin^2 \omega t \cdot d(\omega t) \right]}$$

$$V_{or} = \frac{V_m}{2\sqrt{\pi}} \left[(\pi - \alpha) + \frac{1}{2} \sin 2\alpha \right]^{1/2}$$

- The value of RMS current I_{or} is

$$I_{or} = \frac{V_{or}}{R}$$

- Power delivered to the resistive load = (Rms load voltage)(Rms load current)

$$P = V_{or} \cdot I_{or}$$

$$P = I_{or}^2 R$$

- Input volt amperes = (rms source voltage)(total rms line current) Input VA = $V_s \cdot I_{or}$

$$V_s = \frac{V_m}{\sqrt{2}} \quad \text{and}$$

$$I_{or} = \frac{V_{or}}{R} = \frac{\frac{V_m}{2\sqrt{\pi}} \left[(\pi - \alpha) + \frac{1}{2} \sin 2\alpha \right]^{1/2}}{R}$$

$$= \frac{V_m}{2R\sqrt{\pi}} \left[(\pi - \alpha) + \frac{1}{2} \sin 2\alpha \right]^{1/2}$$

$$I_{or} = \frac{\sqrt{2}V_s}{2R\sqrt{\pi}} \left[(\pi - \alpha) + \frac{1}{2} \sin 2\alpha \right]^{1/2}$$

$$\text{Input VA} = \frac{\sqrt{2}V_s^2}{2R\sqrt{\pi}} \left[(\pi - \alpha) + \frac{1}{2} \sin 2\alpha \right]^{1/2}$$

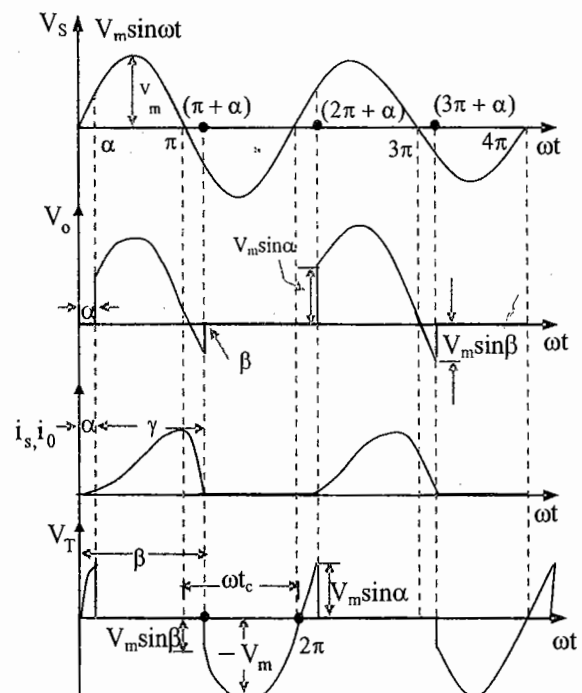
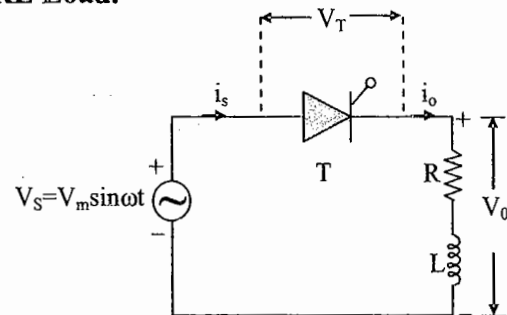
- Input power factor = $\frac{\text{Power delivered to load}}{\text{Input VA}}$

$$= \frac{V_{or} \cdot I_{or}}{V_s \cdot I_{or}} = \frac{V_{or}}{V_s}$$

Input power factor

$$= \frac{1}{\sqrt{2\pi}} \left[(\pi - \alpha) + \frac{1}{2} \sin 2\alpha \right]^{1/2}$$

1 - ϕ Half Wave phase controlled rectifier with RL Load.



- During the positive half cycle of the supply voltage thyristor T is in Forward biased and at $\omega t = \alpha$, thyristor is turned on by gating signal. The load voltage V_o becomes equal to source voltage V_s .
- But the inductance L forces load current ' i_o ' (output current) to rise gradually. After sometime, i_o reaches maximum value and then begins to decrease.
- At $\omega t = \pi$, V_o is zero but ' i_o ' is not zero because of the Load inductance L. After $\omega t = \pi$, SCR is subjected to reverse voltage but it will not be turned off as load current i_o is not less than the holding current. At some angle $\beta > \pi$, i_o reduces to zero and SCR is turned off as it is already reverse biased. After $\omega t = \beta$, $V_o = 0$ and $i_o = 0$.

β = extinction angle and α = firing angle
 $(\beta - \alpha) = \gamma$ is called the conduction angle.

V_T is the voltage across the SCR,

At $\omega t = \alpha$, $V_T = V_m \sin \alpha$

$\omega t = \alpha$ to β , $V_T = 0$

$\omega t = \beta$, $V_T = V_m \sin \beta$

As $\beta > \pi$, I_T is negative at $\omega t = \beta$. Thyristor is there for reverse biased from $\omega t = \beta$ to 2π . The circuit turn-off time is equal to

$$t_c = \frac{2\pi - \beta}{\omega}$$

- Average load voltage

$$V_o = \frac{1}{2\pi} \int_{\alpha}^{\beta} V_m \sin \omega t d(\omega t)$$

$$V_o = \frac{V_m}{2\pi} (\cos \alpha - \cos \beta)$$

- Average load current,

$$I_o = \frac{V_m}{2\pi R} (\cos \alpha - \cos \beta)$$

- rms load Voltage

$$V_{or} = \sqrt{\frac{1}{2\pi} \int_{\alpha}^{\beta} V_m^2 \sin^2(\omega t) d(\omega t)}$$

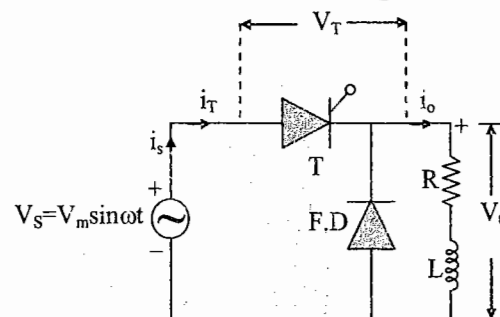
$$V_{or} = \frac{V_m}{2\sqrt{\pi}} \left[(\beta - \alpha) - \frac{1}{2} (\sin 2\beta - \sin 2\alpha) \right]^{1/2}$$

- Input power factor

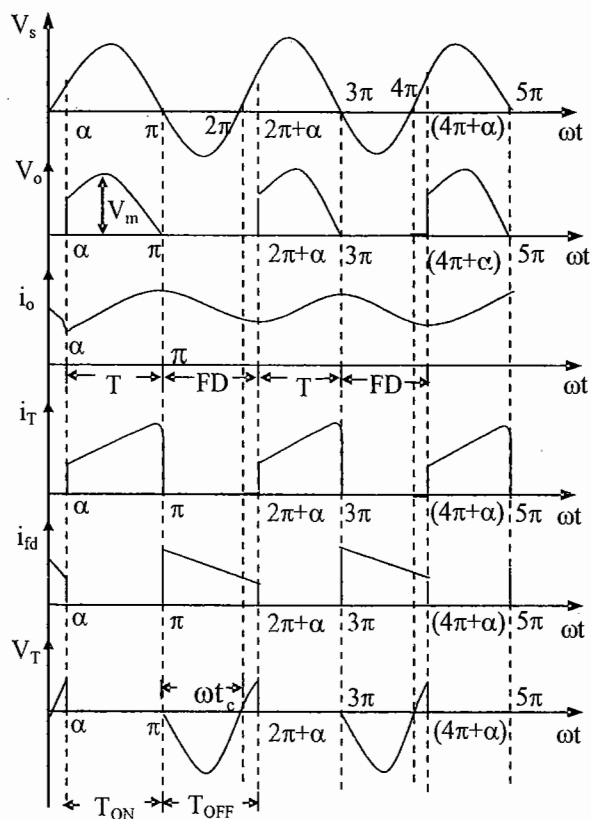
$$= \frac{\text{power delivered to load}}{\text{Input VA}} = \frac{V_{or} I_{or}}{V_s I_{or}}$$

$$\text{Input power factor} = \frac{V_{or}}{V_s}$$

1 - ϕ Half Wave phase controlled rectifier with R - L Load and Free Wheeling Diode



- The waveform of the load current i_o can be improved by connecting a free wheeling diode across load as shown in the above figure.
- A free wheeling diode is also called Bypass or commutating diode.
- At $\omega t = 0$, source voltage is becoming positive, at some delay angle α , forward biased SCR is triggered and source voltage V_s appears across load as V_o .
- At $\omega t = \pi$, source voltage V_s is zero and just after this instant as V_s tends to reverse, freewheeling diode FD is forward biased. Now load current i_o is transferred to FD.
- It is assumed that during free wheeling period, load current does not decay to zero until the SCR is triggered again at $(2\pi + \alpha)$
- Voltage Drop across FD is taken as almost zero, the load voltage ' V_o ' is therefore, during the freewheeling period.



- It is seen from the waveforms that SCR is reversed biased from $\omega t = \pi$ to 2π , therefore circuit turnoff time is

$$t_c = \frac{\pi}{\omega} \text{ sec}$$

- Operation of circuit can be explained in two modes.

(i) Conduction Mode.

(ii) Free wheeling Mode.

SCR Conducts from α to π , $(2\pi + \alpha)$ to 3π hence this mode is called as conduction mode.

In Freewheeling mode SCR is reverse biased and F.D. Conducts from π to $(2\pi + \alpha)$, 3π to $(4\pi + \alpha)$

- Average load voltage

$$V_o = \frac{1}{2\pi} \int_{\alpha}^{\pi} V_m \sin(\omega t) d(\omega t)$$

$$V_o = \frac{V_m}{2\pi} (1 + \cos \alpha)$$

- Average load current, $I_o = \frac{V_o}{R}$

In case of F.D. is not used,

- Load consumes power P_1 from source for α to π (both v_o and i_o are positive) where as energy stored in Inductor L is returned to the source as power P_2 for π to β . (V_o is negative and i_o is positive)
As a result, net power consumed by the load is $P_1 - P_2$.

- If Free wheeling diode is used load absorbs power for α to π , but for π to $(2\pi + \alpha)$, energy stored in 'L' is delivered to load resistance R through the F.D. As a consequence, power consumed by load is more.

- The Advantages by using freewheeling diode are :

(i) input power factor is improved.

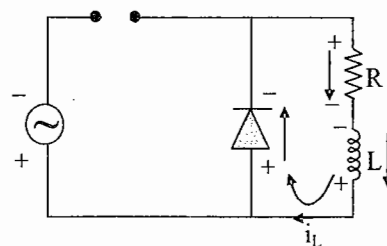
$$\text{Input pf} = \frac{\text{Power delivered to load}}{\text{Input VA}}$$

(ii) load current waveform is improved.

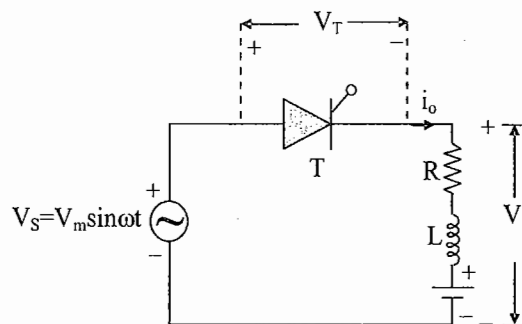
(iii) as a result of (ii) load performance is better and

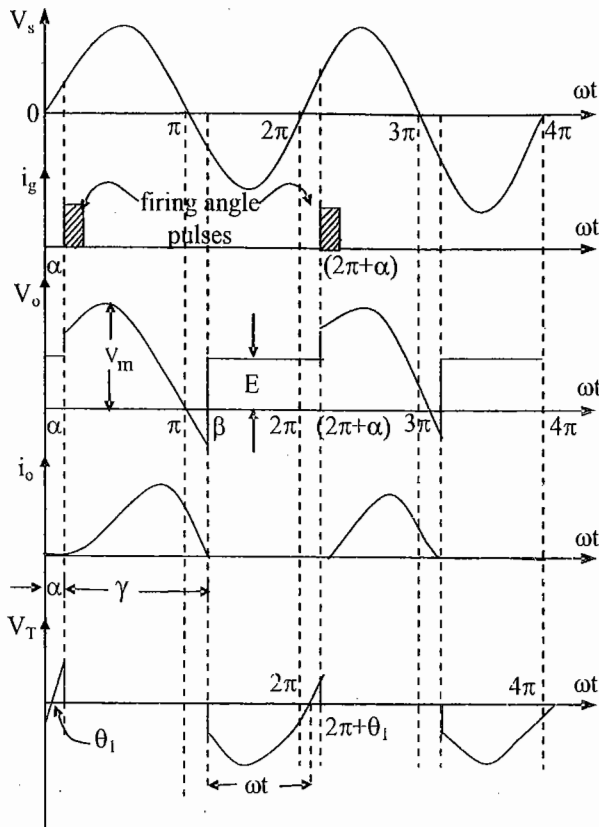
(iv) as energy stored in L is transferred to R during the freewheeling period, over all converter efficiency is improved.

- F.D. prevents the load voltage V_o from becoming negative.



Single Phase (1- ϕ) half wave phase controlled rectifier with RLE Load.





$$\omega t_c = (2\pi + \theta_1 - \beta)$$

- 1-φ half wave rectifier with RLE load is shown above the counter emf E in the load may be due to a battery or a back emf in a DC motor
- The minimum value of firing angle is obtained from the relation $V_m \sin \omega t = E$. This is shown to occur at an angle θ_1 .

$$\theta_1 = \sin^{-1}\left(\frac{E}{V_m}\right)$$

In case SCR is fired at an angle $\alpha < \theta_1$, then $E > V_s$, SCR is reverse biased and therefore it will not turn ON.

- The maximum value of firing angle is $\theta_2 = \pi - \theta_1$.
- The obtained average load voltage
$$V_o = \frac{1}{2\pi} \int_{\alpha}^{\beta} V_m \sin \omega t d(\omega t) + \int_{\beta}^{(2\pi+\alpha)} E d(\omega t)$$

$$V_o = \frac{1}{2\pi} [V_m (\cos \alpha - \cos \beta) + E(2\pi + \alpha - \beta)]$$
- Average value of load current can be obtained by

$$I_o = \frac{1}{2\pi R} \left[\int_{\alpha}^{\beta} (V_m \sin \omega t - E) d(\omega t) \right]$$

$$I_o = \frac{1}{2\pi R} [V_m (\cos \alpha - \cos \beta) - E(\beta - \alpha)]$$

- Rms value of load current, with $L=0$.

$$I_{or} = \sqrt{\frac{1}{2\pi} \int_{\alpha}^{\beta} \left[\frac{V_m \sin \omega t - E}{R} \right]^2 d(\omega t)}$$

- Power delivered to load = P

$$P = I_{or}^2 R + I_o E$$

- Supply power factor

$$= \frac{\text{Power delivered to load}}{\text{Input VA}}$$

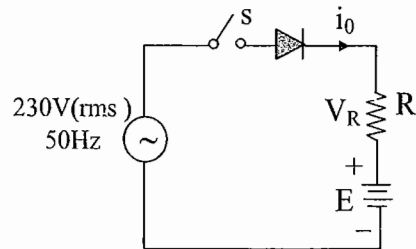
$$\text{Input p.f.} = \frac{I_{or}^2 R + I_o E}{V_s \cdot I_{or}}$$

The circuit turn off time is

$$t_c = \frac{2\pi + \theta_1 - \beta}{\omega}$$

Workedout Example

- A dc battery of constant emf E is being charged through a resistor as shown in below fig. For source voltage of 230V, 50Hz and for $R = 4\Omega$ $E = 100V$. The power dissipated in resistor and charging time of battery respectively are



- (a) 2103W, 0.68h (b) 2603W, 1.36h
(c) 2103W, 1.36h (d) 2603W, 0.68h

$$\text{Sol: } I_{or} = \left[\frac{1}{2\pi} \int_{\theta_1}^{\pi - \theta_1} \left(\frac{V_m \sin \omega t - E}{R} \right)^2 d\omega t \right]^{1/2}$$

$$= \left[\frac{1}{2\pi R^2} \left\{ (V_s^2 + E^2)(\pi - 2\theta_1) + V_s^2 \sin 2\theta_1 - 4V_m E \cos \theta_1 \right\} \right]^{1/2}$$

$$\theta_1 = \sin^{-1}(E/V_m)$$

$$= \sin^{-1}\left(\frac{10}{\sqrt{2} \times 230}\right) = 17.90^\circ$$

$$= \left[\frac{1}{2\pi(16)} \left\{ (230^2 + 100^2)(\pi - 2 \times 17.90 \times \pi/180) + 230^2 \sin(2 \times 17.90^\circ) - 4 \times \sqrt{2} \times 230 \times 100 \cos 17.90^\circ \right\} \right]^{1/2}$$

$$I_{or} = 25.50A$$

$$\text{Power dissipated in resistor} = I_{or}^2 R$$

$$= (25.5)^2 \times 4$$

$$= 2603.7W$$

Average value of current

$$I_0 = \frac{1}{2\pi R} [2V_m \cos \theta_1 - E(\pi - 2\theta_1)]$$

$$= \frac{1}{2\pi \times 4} [2 \times \sqrt{2} \times 230 \times \cos(17.9^\circ) - 100(\pi - 2)(17.9 \times \pi/180)]$$

$$= 14.61A$$

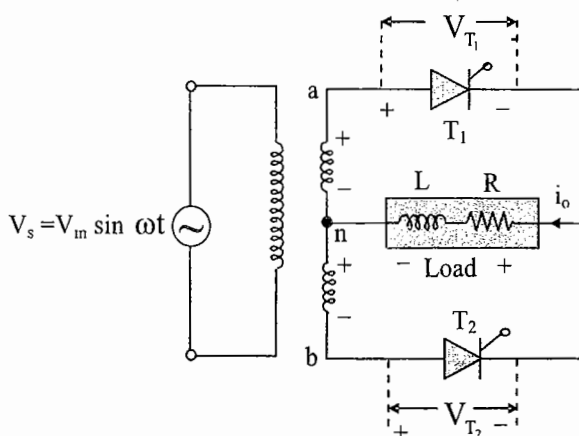
$$\text{Power delivered to battery} = EI_0$$

$$= 100 \times 14.61$$

$$= 1461W$$

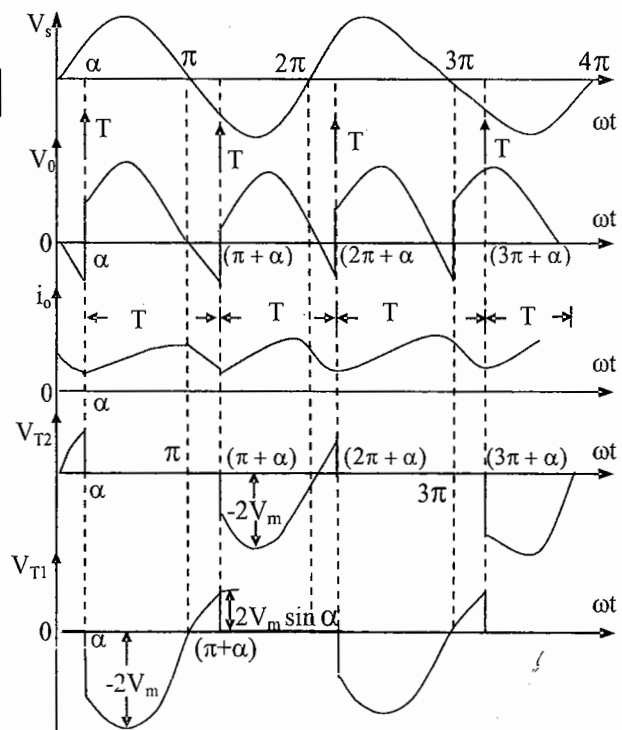
$$\text{Charging time} = \frac{1000}{1461} = 0.684h$$

1 - ϕ Full Wave Phase Controlled Rectifier – Mid Point Converter



- The circuit diagram of a single phase full wave rectifier using a centre tapped transformer.

- When terminal 'a' is positive with respect to 'n', terminal 'n' is positive with respect to b. Therefore, $V_{an} = V_{nb}$ or $V_{an} = -V_{bn}$, as 'n' is the mid point of secondary winding.
- It is assumed here that turns ratio from primary to each secondary is unity.
- Line Commutation: The process of SCR turn off by natural reversal of ac supply voltage is called natural (or) line commutation.
- At $\omega t = \alpha$, T_1 is triggered, and SCR1 conducts from α to $(\pi + \alpha)$. After $\omega t = \pi$, T_1 is reverse biased but it will continue conducting because forward biased SCR T_2 is not yet triggered. At $\omega t = \pi + \alpha$, T_2 is triggered. T_1 is reverse biased by voltage of magnitude $2V_m \sin \alpha$, current is transferred from T_1 to T_2 , T_1 is therefore turned off.



- It is assumed here that load or output current is continuous.
- The turn – off time provided by this circuit is $t_c = \frac{\pi - \alpha}{\omega}$
- It is seen from voltage waveform V_o , average value of output voltage is given by

$$V_o = \frac{1}{\pi} \int_{\alpha}^{\pi+\alpha} V_m \sin \omega t d(\omega t)$$

$$V_o = \frac{2V_m}{\pi} \cos \alpha$$

for only R – load it is, $V_o = \frac{V_m}{\pi} [1 + \cos \alpha]$

- RMS load voltage

$$= V_{or} = \sqrt{\frac{1}{\pi} \int_{\alpha}^{\pi+\alpha} V_m^2 \sin^2 \omega t d(\omega t)}$$

$$V_{or} = \frac{V_m}{\sqrt{2}} = V_s$$

- Average Load current $I_o = \frac{V_o}{R}$
- Value of RMS current $I_{or} = \frac{V_{or}}{R}$
- Input power factor

$$= \frac{\text{Power delivered to Load}}{\text{Input VA}} = \frac{V_{or}}{V_s}$$

Disadvantages:

- Peak inverse voltage (PIV) of the SCR is two times of the maximum value of the supply voltage.
- This circuit requires Mid point transformers which is costly.

Worked out Examples

- 02 A single – phase mid point SCR converter supplies constant load current of 5 A when the triggering angle is maintained at 35° . The input voltage to the converter is 220V, 50 Hz. The turns ratio from primary to each secondary is 1/2. Determine the load voltage.

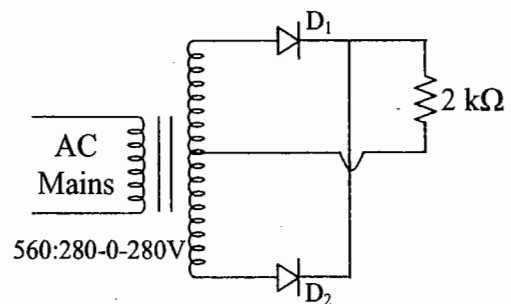
$$\text{Sol: } V = \frac{220}{(1/2)} = 440 \text{ V}$$

$$I_o = 5 \text{ A}$$

$$V_o = \frac{2V_m}{\pi} \cos \alpha$$

$$= \frac{2 \times 440 \times \sqrt{2}}{\pi} \cos 35^\circ = 324.4 \text{ V}$$

03. The center-tap full-wave single-phase rectifier circuit uses 2 diodes as shown in the given figure. The rms voltage across each diode is



(a) 790.7 V

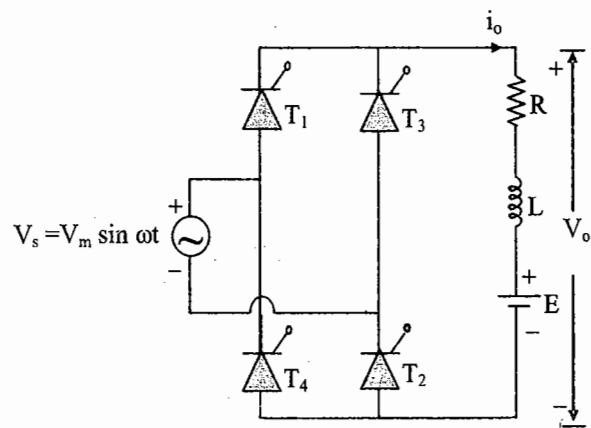
(b) 395.3 V

(c) 280 V

(d) 201.3 V

$$\text{Sol: } V_{Drms} = \frac{2V_m}{2} = 280\sqrt{2}$$

1 - ϕ Full Wave Bridge Phase Controlled Rectifier with RLE Load

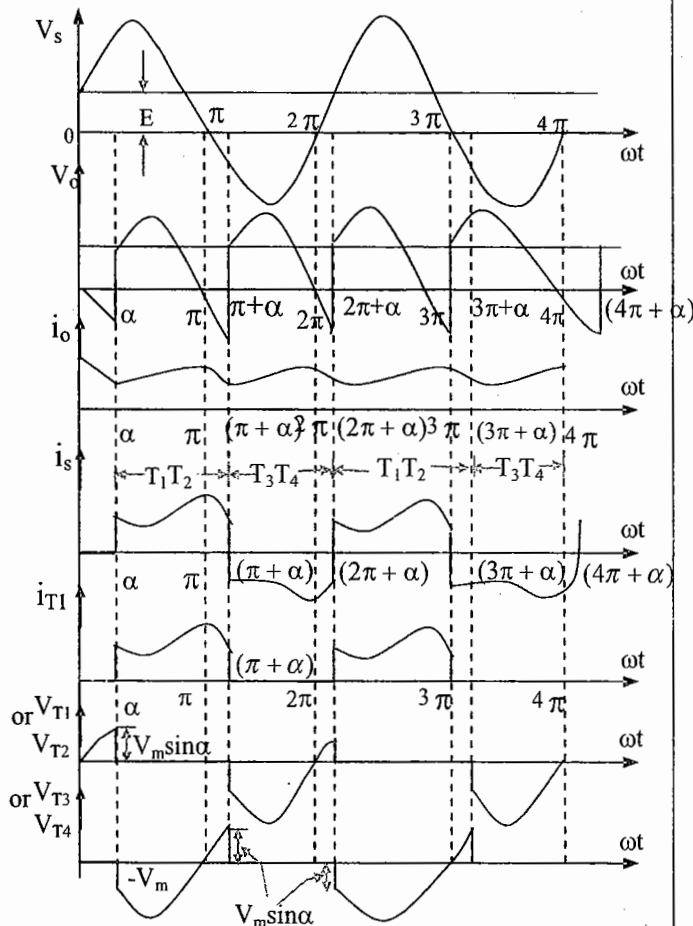


- Thyristor pair T_1, T_2 are simultaneously triggered and after π radians T_3, T_4 are triggered.
- During positive half cycle of the supply i.e. $\omega t = 0$ to π , T_1, T_2 are forward biased, at firing angle α they are triggered.
- Note that when T_1, T_2 are gated at $\omega t = \alpha$, these SCRs will get turned on only if



$V_m \sin \alpha > E$. Thyristors T_1, T_2 conduct from $\omega t = \alpha$ to $(\pi + \alpha)$.

- At $\omega t = (\pi + \alpha)$, Forward biased SCR's T_3, T_4 are triggered. The supply voltage turns off T_1, T_2 by natural commutation and the load current is transferred from T_1, T_2 to T_3, T_4 .



- The maximum reverse voltage across T_1, T_2, T_3 or T_4 is V_m and at the instant of triggering with firing angle α , such SCR is subjected to a reverse voltage of $V_m \sin \alpha$.
- The load terminal voltage, or full-bridge converter output voltage, V_o .

The average value of output voltage V_o is given by.

$$V_o = \frac{1}{\pi} \int_{\alpha}^{(\pi+\alpha)} V_m \sin \omega t d(\omega t)$$

$$= \frac{2V_m}{\pi} \cos \alpha$$

$$V_o = \frac{2V_m}{\pi} \cos \alpha$$

- Rms value of output voltage is.

$$V_{or} = \sqrt{\frac{1}{\pi} \int_{\alpha}^{(\pi+\alpha)} V_m^2 \sin^2 \omega t d(\omega t)}$$

$$= \frac{V_m}{\sqrt{2}} = V_s$$

- The advantages of 1- ϕ Bridge converter over 1- ϕ mid point converter are.
 - SCRs are subjected to a peak inverse voltage of $2V_m$ in mid point converter and V_m in Bridge converter. Thus for the same voltage and current ratings of SCRs, power handled by Bridge configuration is about half of that of the midpoint configuration.
 - In Mid-point converter, each secondary should be able to supply the load power. As such, the transformer rating in mid-point converter is double the load rating. This can be avoided in 1- ϕ Bridge converter.

Worked out Examples

04. A single-phase full converter feeding RLE load has the following data source voltage = 230V, 50 Hz, $R = 2.5\Omega$, $E = 100V$ and firing angle = 30° . If load inductance is large enough to make the load current virtually constant, then

- Compute the average value of load voltage and load current.
- Compute the input pf.

Sol: (a) $V_o = \frac{2V_m}{\pi} \cos \alpha$

$$= \frac{2 \times 230 \sqrt{2}}{\pi} \cos 30^\circ = 179.3 V$$

$$I_o = \frac{V_o - E}{R} = \frac{179.3 - 100}{2.5} = 31.72 A$$

- Power factor = $0.9 \cos \alpha$



$$= 0.9 \cos 30^\circ = 0.78 \text{ lag}$$

05. A single phase full converter feeds power to RLE load with $R=10\Omega$, $L=10\text{mH}$, and $E=50\text{V}$, the ac source voltage is 230V , 50Hz . For continuous conduction, what is the average value of load current for firing angle delays of 60° ? (IES - 09)

- (a) 4.63 A (b) 6A
(c) 6.5 A (d) 5.35 A

Sol: Given $R=10\Omega$, $L=10\text{mH}$
And $E=50\text{V}$.

$$V_S = 230\text{V}, \alpha = 60^\circ I_{\text{avg}} = ?$$

In single phase full converter with RLE load

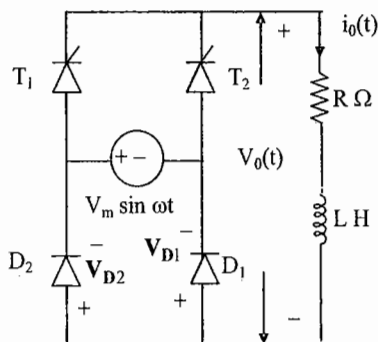
$$V_0 = E + I_0 R.$$

$$\frac{2V_m}{\pi} \cos \alpha = E + I_0 R$$

$$\frac{2 \times \sqrt{2} \times 230}{\pi} \cos 60^\circ = 50 + I_0(10)$$

$$I_0 = 5.35\text{A}$$

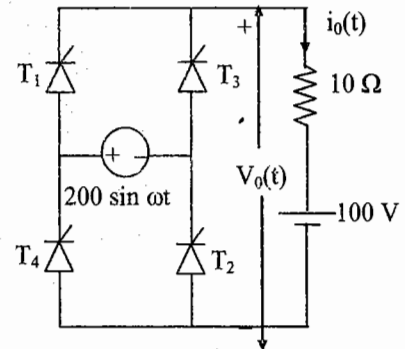
06. In the circuit of figure, (1). At $\omega t = \pi$, $v_0(t) = \underline{\hspace{2cm}}$. $i_0(t) = \underline{\hspace{2cm}}$. (2) for $\omega t > \pi$, till $i_0(t)$ reaches zero, if T_1 continues to conduct, it is diode which conducts along with it; and the load voltage is . The blanks are, respectively.



- (a) 0, > 0 , D_2 0
(b) > 0 , equals to zero, D_1 , $V_m \sin \omega t$
(c) 0, < 0 , D_2 , $-V_m \sin \omega t$
(d) > 0 , 0, D_1 , 0

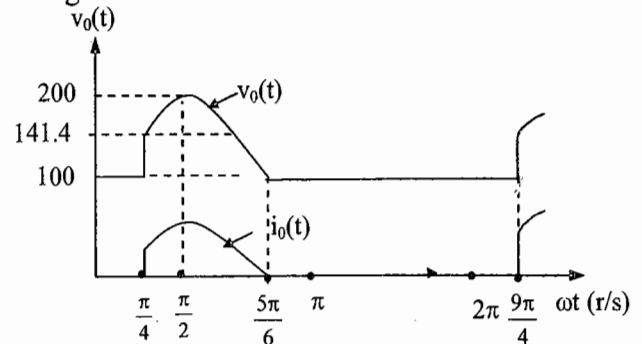
Sol: At $\omega t = \pi$; the supply voltage goes through zero and starts becoming negative. Because of the inductance, the load current i_0 still remains positive (> 0). If this current flows through D_1 , remains zero and V_{D2} must become +ve. This is not allowed by the diode. So D_2 conducts, becomes zero and $V_{D1} = -V_m \sin \omega t$. D_1 is off. Since T_1 & D_2 are on, $v_0(t)$ becomes zero.

07. In figure, thyristor T_3 is damaged and is permanently open. T_1 and T_2 are triggered at $\omega t = \left(\frac{\pi}{4}, \frac{9\pi}{4}, \dots\right)$ and T_3 and T_4 are triggered at $\omega t = \left(\frac{5\pi}{4}, \frac{13\pi}{4}, \dots\right)$. Average value of $i_0(t)$ (in amp) is



- (a) 10 (b) 41.4
(c) 4.14 (d) 6.78

Sol: Waveforms of $v_0(t)$ and $i_0(t)$ are shown in figure.

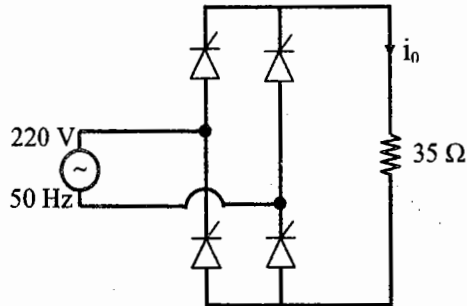


$$i_0(t) = \begin{cases} \frac{200 \sin \omega t - 100}{10} \text{ A} & \left[\frac{\pi}{4} \leq \omega t \leq \frac{5\pi}{6} \right] \\ 0 & \text{elsewhere} \end{cases}$$

$$I_{0_{arc}} = \frac{1}{2\pi} \int_{\frac{\pi}{4}}^{\frac{5\pi}{6}} (20 \sin \omega t - 10) d(\omega t)$$

$$= 6.78 \text{ A}$$

08. A 1- ϕ fully controlled rectifier is supplying power to a resistive load of 35Ω as shown in figure. If the bridge is triggered at 90° , the average current drawn by the load is given by

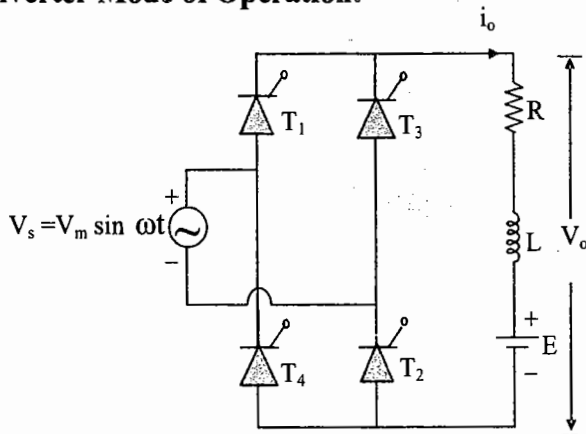


- (a) 0 A (b) $\sqrt{2}$ A
(c) $2\sqrt{2}$ A (d) $\frac{1}{\sqrt{2}}$ A

Sol: For $\alpha = 90^\circ \Rightarrow V_0 = \frac{V_m}{\pi}$

$$I_0 = \frac{V_m}{\pi R} = 2\sqrt{2} \text{ A}$$

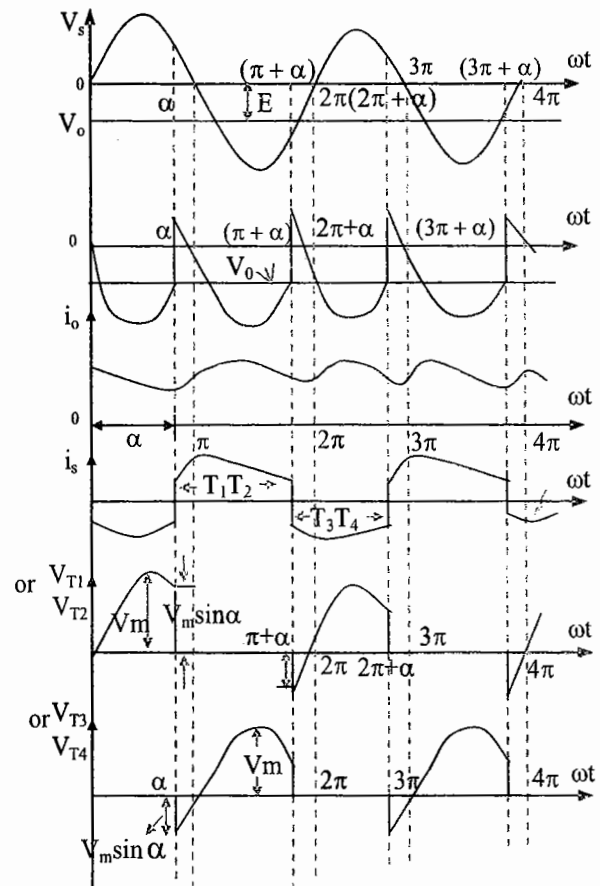
Inverter Mode of Operation:



- If the load circuit emf E is reversed and with $\alpha > 90^\circ$; then this dc source E will feed power back to ac source.
- This mode of operation of a converter with $\alpha > 90^\circ$ is known as inverter operation.
- If $\alpha > 90^\circ$, average output voltage is negative, current is positive so the average power would

be negative. The net power flows from D.C load to A.C source. This operation is known as Line commutated inverter.

- Regenerative braking employs the principle of Line commutated inverter.
- During zero to α , ac source voltage V_s is positive but ac source current is negative, power therefore power flows from dc source to ac source. From α to π , both V_s and I_s are positive, therefore power flows from ac source to dc source. But the net power flow is from AC source to DC source.
- In converter operation, the average value of output voltage V_0 must be greater than load circuit emf E . During inverter operation, Load circuit emf must be more than ac supply voltage.
- In both converter and inverter modes, thyristor must be forward biased and current through SCRs must flow in the same direction as these are unidirectional devices. This is the reason output current i_o shown positive.



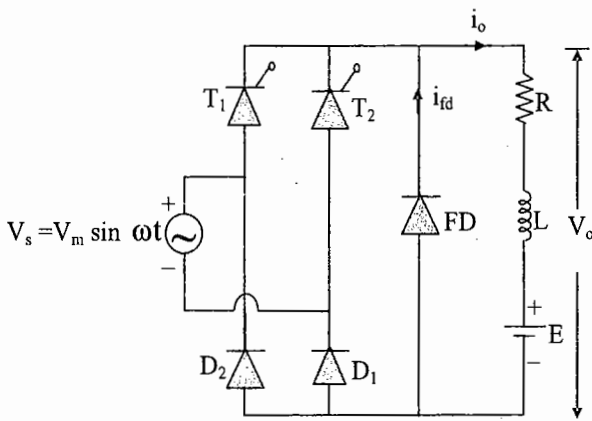
- The circuit turn off time for both converter and inverter operations is given by

$$t_c = \frac{\pi - \alpha}{\omega} \text{ sec}$$

- Following conditions are to be satisfied to achieve Regenerative braking.
 - $\alpha > 90^\circ$
 - $E = \text{Negative}$
 - $E > V^0$
- E in the motor can be made negative either by making the field flux to be negative (or) speed of rotation to be negative.

1- ϕ Full Wave Semi Phase Controlled Rectifier with RLE Load: (or)

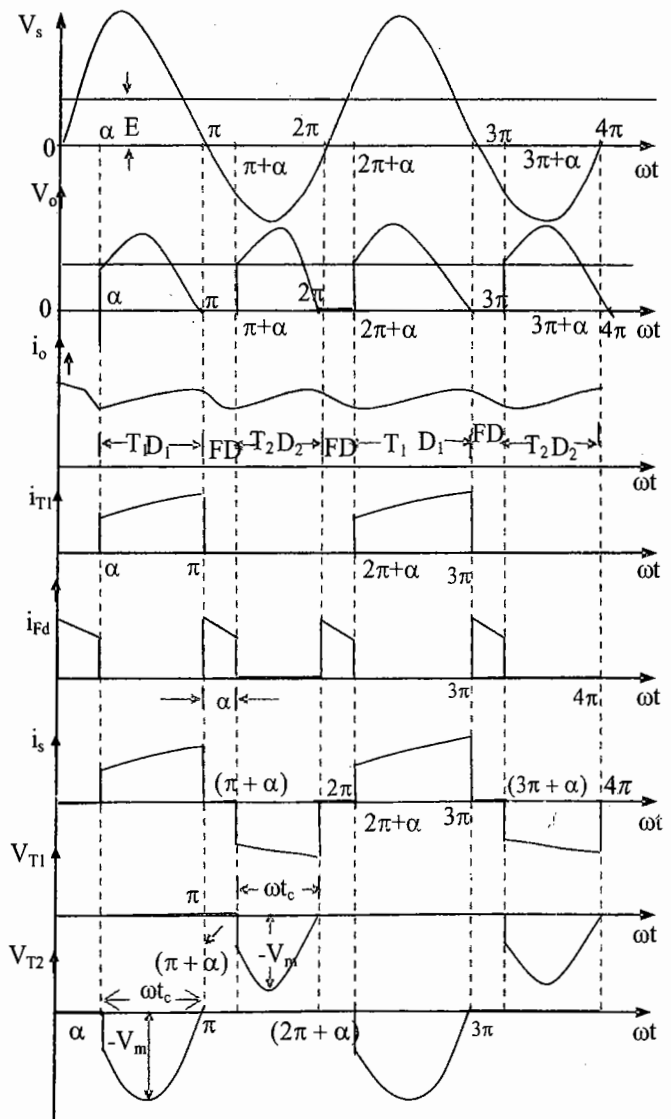
1- ϕ Full Wave Semi Controlled Rectifier with Freewheeling Diode:



- After $\omega t = 0$, thyristor T_1 is forward biased only when source voltage $V_m \sin \omega t$ exceeds E . thus, T_1 is triggered at a firing angle α such that $V_m \sin \alpha > E$.
- With T_1 on load gets connected to source through T_1 and D_1 . Load current i_o flows through $RLE \rightarrow D_1 \rightarrow V_s \rightarrow T_1 \rightarrow RLE$.
- As SCR T_1 is reverse biased the out put current i_o flows through freewheeling diode.
- During negative half cycle of the source, T_2 will be forward biased only when source voltage is more than E . At $\omega t = \pi + \alpha$, source voltage exceeds E , T_2 is therefore triggered.

Output current flows through $V_s \rightarrow T_2 \rightarrow RLE \rightarrow D_2 \rightarrow V_s$.

- After $\omega t = 2\pi$, FD is again forward biased and output current freewheels through FD.
- During the interval α to π , T_1 and D_1 conduct and the energy deliver by ac source is partially stored in inductor L , partially stored as electrical energy in load circuit emf E and partially dissipated as heat in R . During the free wheeling period π to $(\pi + \alpha)$, energy stored in inductor is recovered and is partially dissipated in R and partially added to the energy stored in load emf E .
- No energy is fed back to the source during free wheeling period.



- The average output voltage = V_o .

$$V_o = \frac{1}{\pi} \int_{\alpha}^{\pi} V_m \sin \omega t \, d(\omega t)$$

$$V_o = \frac{V_m}{\pi} (1 + \cos \alpha)$$

- Rms value of output voltage is

$$V_{or} = \sqrt{\frac{1}{\pi} \int_{\alpha}^{\pi} V_m^2 \sin^2 (\omega t) \, d(\omega t)}$$

$$V_{or} = V_s \sqrt{\left[\frac{1}{\pi} \left\{ (\pi - \alpha) + \frac{\sin 2\alpha}{2} \right\} \right]}$$

- Comparison between 1 - ϕ Full converter and 1 - ϕ semi converter regarding to average output voltage (V_o).

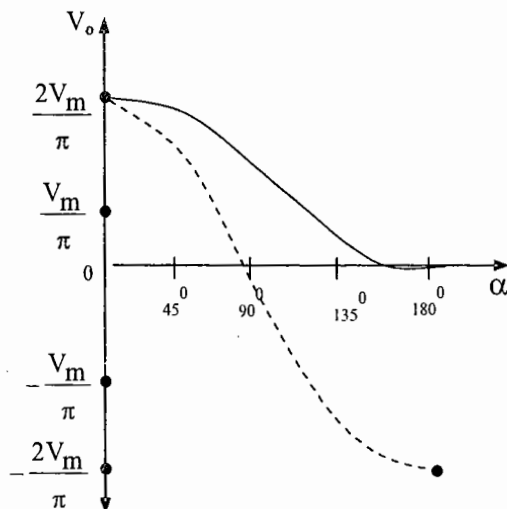
V_o of full converter is ;

$$V_{or} = V_s \sqrt{\left[\frac{1}{\pi} \left\{ (\pi - \alpha) + \frac{\sin 2\alpha}{2} \right\} \right]}$$

V_o of semi converter is;

$$V_o = \frac{V_m}{\pi} (1 + \cos \alpha)$$

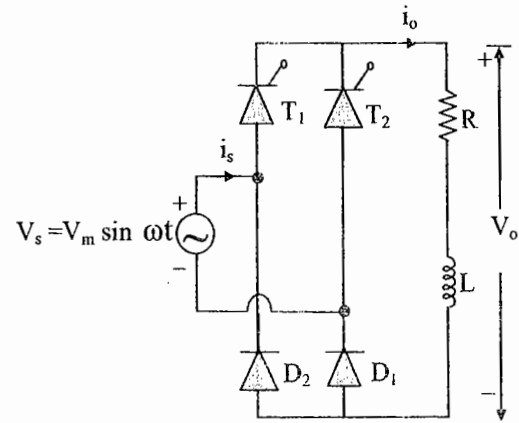
The graph between output voltage of the Rectifier and firing angle is.



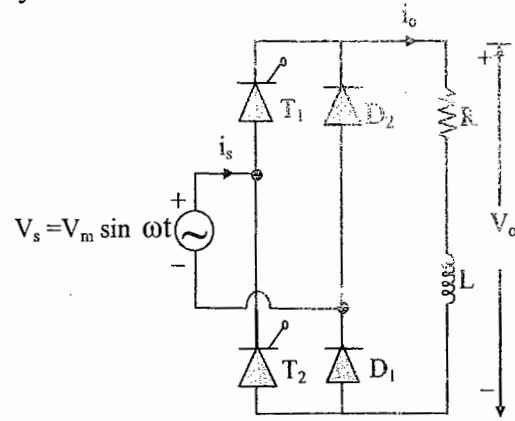
- The circuit turn off time of the semi converter is $t_c = \frac{\pi - \alpha}{\omega}$ (sec).
- A 1 - ϕ semi converter, employing one SCR and one diode in each leg, is called single

phase symmetrical semi converter. (or) 1 - ϕ two pulse Symmetrical semi converter.

- It is also called as 1 - ϕ half controlled symmetrical converter.



- If two SCRs in one leg and two diodes in the other leg, is called single phase asymmetrical semi converter. (or) 1 - ϕ two pulse asymmetrical semi converter.
- It is also called as 1 - ϕ Half controlled asymmetrical converter.



Advantages of 1 - ϕ Semi Phase Controlled Rectifier over 1 - ϕ Full Converter:

- Higher value of dc output voltage.
- 1 - ϕ semi converter required two SCRs and three (or two) diodes but a 1 - ϕ full converter needs four SCRs. 1 - ϕ semi converter circuits are therefore, cheaper.
- Freewheeling action in semi converter circuits render better power factor than its value in full converter circuits.



- Performance parameters of semi converter circuits are superior than their corresponding values in full converter circuits.
- Disadvantage of 1 - ϕ semi converter is it offers only one quadrant operation, where as 1 - ϕ full converter can furnish two quadrant operation.

Worked out Examples

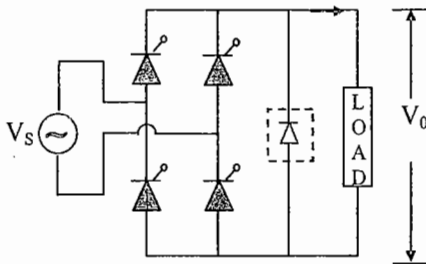
09. A single phase full bridge converter with a free wheeling diode feeds an inductive load. The load resistance is 15.53Ω and it has a large inductance providing constant and ripple free d.c. current. Input to converter is fed from an ideal 230V, 50Hz single phase source. For a firing angle delay of 60° , the average value of diode current is

(IES - 02)

- (a) 10A (b) 8.165 A
(c) 5.774 A (d) 3.33 A

Sol: $R_L = 15.53 \Omega$, $\alpha = 60^\circ$

$V_s = 230V$, 50Hz $I_{F.D} = ?$



A full wave rectifier with F.D acts as a semi converter.

Average output voltage (V_o)

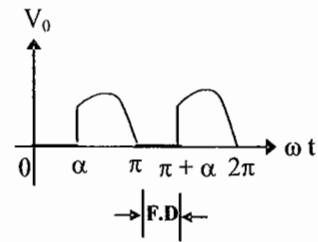
$$= \frac{V_m}{\pi} (1 + \cos \alpha)$$

Average output current

$$I_o = \frac{V_o}{R}$$

$$I_o = \frac{\sqrt{2} \times V_s}{\pi \times 15.53} (1 + \cos 60^\circ)$$

$$= \frac{\sqrt{2} \times 230 \times 1.5}{\pi \times 15.53} = 10A$$



Assume output current is constant at $I_o = 10A$. Freewheeling diode conducts for α duration in every half cycle. Average current through F.D

$$= I_o \times \frac{\alpha}{\pi} = 10 \times \frac{60}{180} = 3.33A$$

10. A single phase two-pulse diode rectifier has input supply of 230V, 50Hz and the load resistance $R = 600\Omega$. Calculate the value of inductance to be connected in series with R so as to limit the current ripple factor to 4%.

- (a) 21.11H (b) 12.11H
(c) 1.211H (d) 11.21H

Sol: current ripple factor

$$CRF = 0.4715 \cdot \frac{R}{\sqrt{R^2 + (2\omega L)^2}}$$

$$0.04 = 0.4715 \cdot \frac{600}{\sqrt{(600)^2 + (2 \times 2\pi \times 50 \times L)^2}}$$

$$L = 11.21H$$

11. A diode whose internal resistance 10Ω is to supply power to a 2000Ω load from a 230V(rms) source of supply calculate the percentage regulation from no load to given load

- (a) +1% (b) -1%
(c) 0.5% (d) -0.5%

Sol: No load voltage $V_o = \frac{1}{2\pi} \int_0^\pi V_m \sin \omega t d\omega t$

$$= \frac{V_m}{2\pi} [-\cos \omega t]_0^\pi$$

$$= \frac{V_m}{\pi}$$

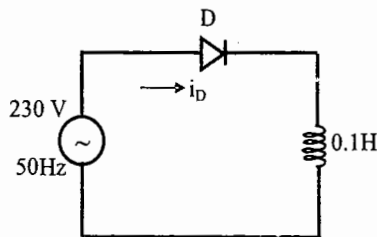
$$= \frac{\sqrt{2} \times 230}{\pi} = 103.521V$$



$$\begin{aligned}\text{At given load } V_L &= \frac{V_m}{\pi} \times \frac{2000}{2010} \\ &= 103.005\end{aligned}$$

$$\begin{aligned}\text{Voltage regulation} &= \frac{V_0 - V_L}{V_0} \times 100 \\ &= \frac{103.521 - 103.005}{103.521} \\ &= 0.498 \\ &\approx 0.5\%\end{aligned}$$

12. In the following single-phase diode Rectifier circuit, the average and rms current rating of the diode will be respectively



- (a) $I_{D_{av}} = 10.35$ A and $I_{D_r} = 14.64$ A
(b) $I_{D_{av}} = 14.64$ A and $I_{D_r} = 10.35$ A
(c) $I_{D_{av}} = 7.32$ A and $I_{D_r} = 10.35$ A
(d) $I_{D_{av}} = 10.35$ A and $I_{D_r} = 12.676$ A

Sol: $I_{0 \max} = \frac{2V_m}{\omega L} = \frac{2 \times 230\sqrt{2}}{100\pi \times 0.1} = 20.7$ A

$$I_{D_{av}} = \frac{I_{D_{\max}}}{2} = 10.35 \text{ A}$$

$$I_{D_{rms}} = I_{D_{av}} \sqrt{\frac{3}{2}} = 12.676 \text{ A}$$

13. In a single phase semi converter, THD in source current is found to be 31%. Then firing angle could be

- (a) 30° (b) 40°
(c) 45° (d) 60°

Sol: If THD = 31%
 $\Rightarrow$ pulse width of $i_A = 120^\circ$ i.e. $\alpha = 60^\circ$

Common Data for Q. No. 14 & 15

A single phase semi converter is operated from 230V, 50 Hz AC supply and with a firing angle of

$\frac{\pi}{4}$. The load on the converter is highly inductive with a resistance of 17.765 ohm and load current is ripple free

14. The RMS value of the freewheeling current is
(a) 10 A (b) 3.33 A
(c) 5 A (d) None
15. RMS value of fundamental source current will be
(a) 10 A (b) 9 A
(c) 5 A (d) 8.315 A

Sol:

$$14. I_0 = \frac{V_0}{R} = \frac{\frac{230\sqrt{2}}{\pi} \left[1 + \cos \frac{\pi}{4} \right]}{17.765} = 10 \text{ A}$$

$$I_{FD \text{ rms}} = I_0 \sqrt{\frac{\alpha}{\pi}} = 5 \text{ A}$$

$$\begin{aligned}15. I_{s1} &= \frac{2\sqrt{2}}{\pi} I_0 \cos \frac{\alpha}{2} \\ &= 0.9 \times 10 \times \cos \frac{\pi}{2} = 8.135 \text{ A}\end{aligned}$$

16. (a) A half controlled single-phase bridge rectifier is supplying an RL Load. It is operated at a firing angle α and the load current is continuous. The fraction of cycle that the free wheeling diode conducts is ---
(GATE-12)

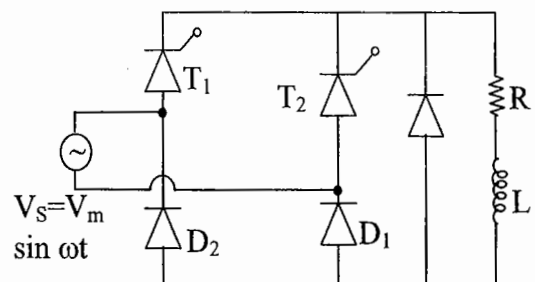
- (a) $1/2$ (b) $(1-\alpha/\pi)$
(c) $\alpha/2\pi$ (d) α/π

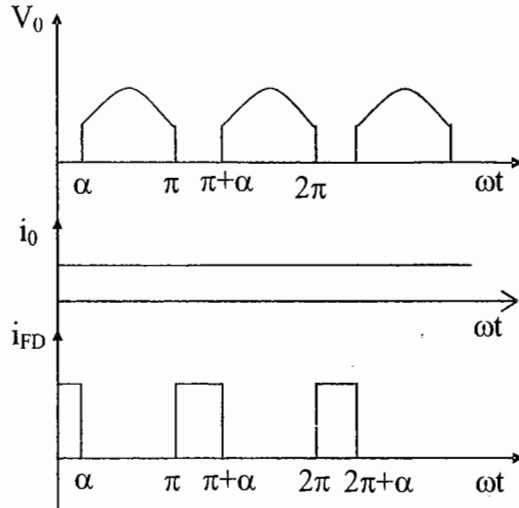
- (b) If it is 3 - ϕ half controlled rectifier, then free wheeling duration is

Ans: (d)

Sol:

(a)





Conduction of Freewheeling diode in a cycle
= 2α

Fraction of conduction time of freewheeling
diode in cycle = $\frac{2\alpha}{2\pi}$

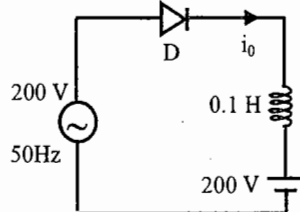
$$\therefore t = \frac{\alpha}{\pi}$$

(b) Free wheeling duration = 0 when $\alpha < 60^\circ$

$$= \frac{\left(\alpha - \frac{\pi}{3}\right) 3}{2\pi} \text{ when } \alpha > 60^\circ$$

17. In the following rectifier circuit, the load current will be maximum at $\omega t =$

- (a) 90°
(b) 135°
(c) 45°
(d) Load current does not have max



3 - ϕ Half Wave Converter:

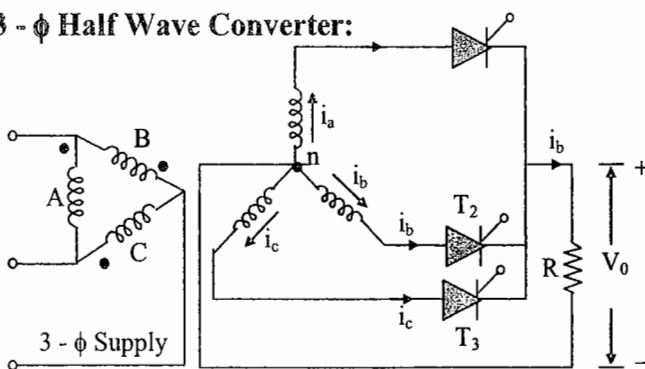
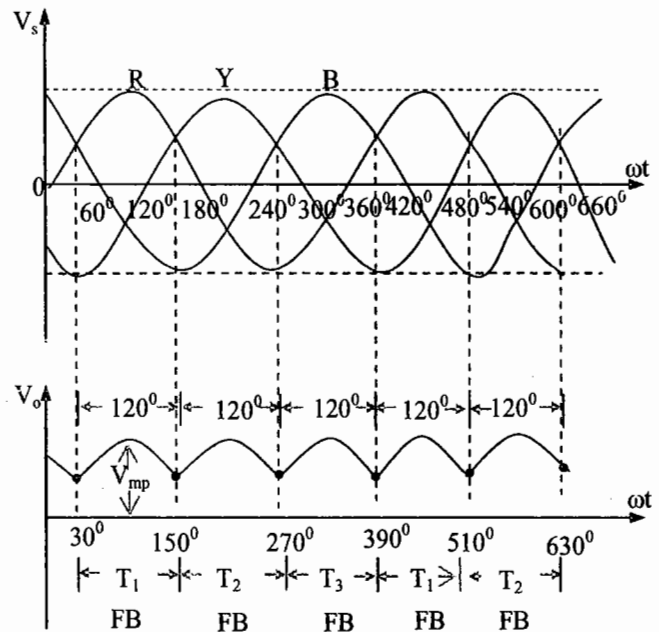
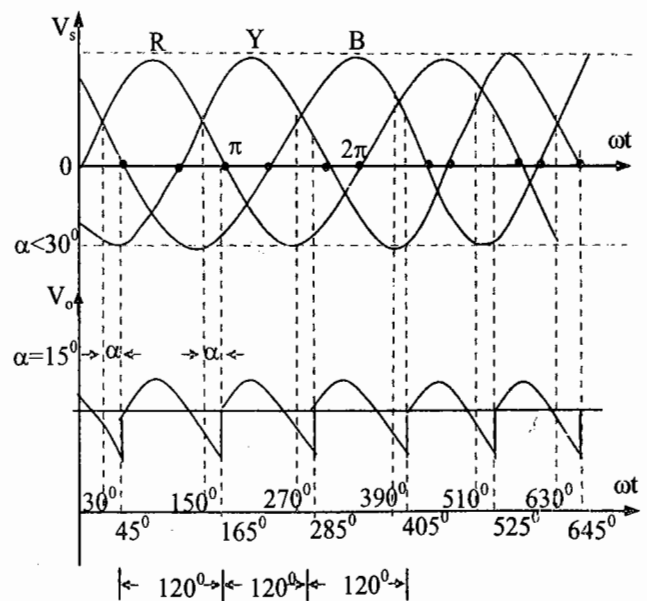


Fig 3 ϕ half wave converter feeding or load



- If firing angle is zero degrees, SCR T_1 would begin conducting from $\omega t = 30^\circ$ to 150° . T_2 from $\omega t = 150^\circ$ to $\omega t = 270^\circ$ and T_3 from $\omega t = 270^\circ$ to 390° . Operation of this converter is now described for $\alpha < 30^\circ$ and for $\alpha > 30^\circ$.



- Firing angle = $\alpha < 30^\circ$.
The output voltage waveform V_o , for firing angle less than 30° ($\alpha = 15^\circ$) is sketched. Where T_1 conducts from $\omega t = (30^\circ + \alpha)$ to



$\omega t = (150^\circ + \alpha)$ and T_2 from $(150^\circ + \alpha)$ to $(270^\circ + \alpha)$ and soon. Each SCR conducts for 120° .

- Average value of output voltage,

$$V_o = \frac{3}{2\pi} \int_{\pi/6+\alpha}^{5\pi/6+\alpha} V_{mp} \sin \omega t d(\omega t)$$

$$= \frac{3V_{mp}}{2\pi} \left(\cos \left(\frac{\pi}{6} + \alpha \right) - \cos \left(\frac{5\pi}{6} + \alpha \right) \right)$$

$$= \frac{3V_{mp}}{2\pi} \sin \left(\frac{\pi}{2} + \alpha \right) \sin \left(\frac{\pi}{3} \right)$$

$$V_o = \frac{\sqrt{3}V_{mp}}{2\pi} \cos \alpha \Rightarrow V_o = \frac{3V_{mL}}{2\pi} \cos \alpha$$

Where, V_{mp} = Maximum value of phase (line to neutral) voltage.

V_{mL} = Maximum value of line

voltage = $\sqrt{3} V_{mp}$.

- Average load current, $I_o = \frac{V_o}{R}$

$$I_o = \frac{3 V_{mL}}{2\pi R} \cos \alpha$$

- Rms value of output, or load voltage is.

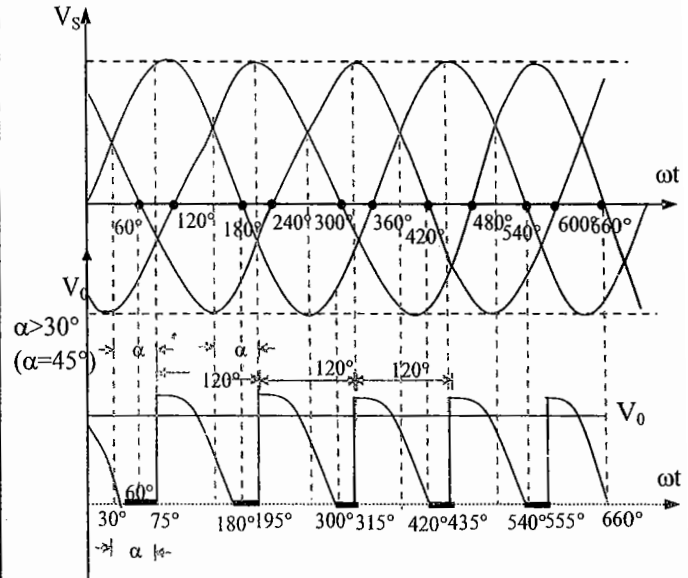
$$V_{or} = \left[\frac{3}{2\pi} \int_{\alpha+\pi/6}^{\alpha+5\pi/6} V_{mp}^2 \sin^2 \omega t d(\omega t) \right]^{1/2}$$

$$V_{or} = V_{mL} \left[\frac{1}{6} + \frac{\sqrt{3}}{8\pi} \cos 2\alpha \right]^{1/2}$$

- Rms load current,

$$I_{or} = \frac{V_{or}}{R} = \frac{V_{mL}}{R} \left[\frac{1}{6} + \frac{\sqrt{3}}{8\pi} \cos 2\alpha \right]^{1/2}$$

- Firing angle $\alpha > 30^\circ$ (say $\alpha = 45^\circ$)
When firing angle is more than 30° , T_1 would conduct from $30^\circ + \alpha$ to 180° , T_2 from $(150^\circ + \alpha)$ to 300° and so on.



Average value of load voltage,

$$V_o = \frac{3}{2\pi} \int_{\alpha+\pi/6}^{\pi} V_{mp} \sin \omega t d(\omega t)$$

$$= \frac{3V_{mp}}{2\pi} \left(-\cos \omega t \right) \Big|_{\alpha+\pi/6}^{\pi}$$

$$V_o = \frac{3V_{mp}}{2\pi} (1 + \cos (\alpha + 30^\circ))$$

- Rms value of output voltage,

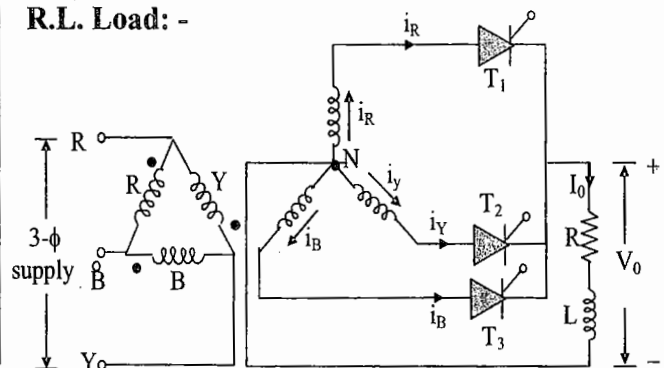
$$V_{or} = \sqrt{\frac{3}{2\pi} \int_{\alpha+\pi/6}^{\pi} V_{mp}^2 \sin^2 \omega t d(\omega t)}$$

$$= \sqrt{\frac{3V_{mp}^2}{4\pi} \left[\left(\frac{5\pi}{6} + \alpha \right) + \frac{1}{2} \left(\sin \frac{2\pi}{6} + 2\alpha \right) \right]}$$

$$V_{or} = \frac{\sqrt{3} \cdot V_{mp}}{2\sqrt{\pi}} \left[\left(\frac{5\pi}{6} - \alpha \right) + \frac{1}{2} \sin \left(2\alpha + \frac{\pi}{3} \right) \right]^{1/2}$$

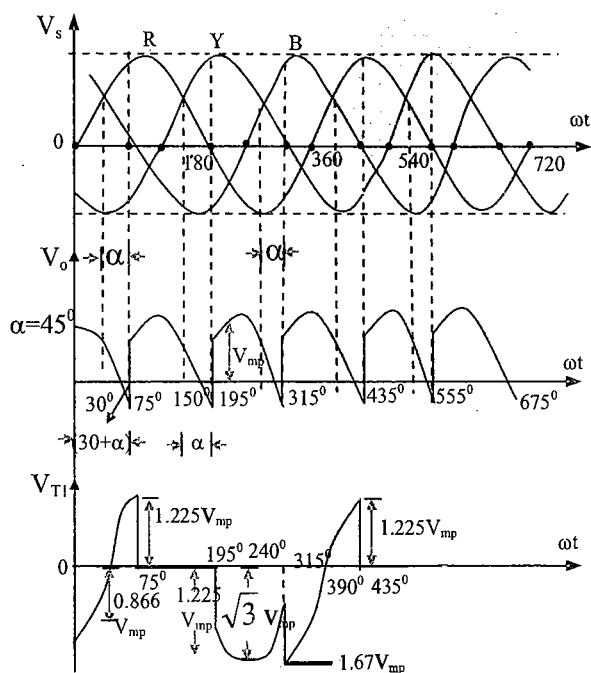
$$V_{or} = \frac{V_{mL}}{2\sqrt{\pi}} \left[\left(\frac{5\pi}{6} - \alpha \right) + \frac{1}{2} \sin \left(2\alpha + \frac{\pi}{3} \right) \right]^{1/2}$$

3 - ϕ Half wave Phase Controlled Rectifier with R.L. Load: -



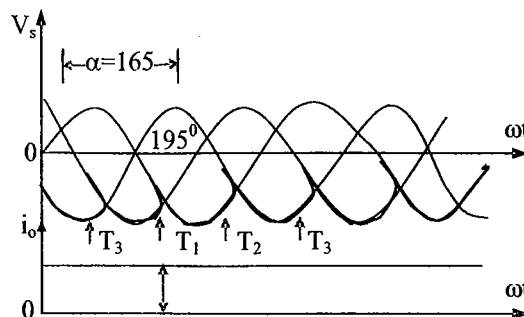


- For firing angles of (i) $30^\circ < \alpha < 90^\circ$ and (ii) $90^\circ < \alpha < 180^\circ$, this converter behaves differently.
- $\alpha = 45^\circ$ say ($30^\circ < \alpha < 90^\circ$)
 T_1 conducts from $(30^\circ + \alpha)$ to $(150^\circ + \alpha)$
 T_2 conducts from $(150^\circ + \alpha)$ to $(270^\circ + \alpha)$
 T_3 conducts from $(270^\circ + \alpha)$ to $(390^\circ + \alpha)$
- At $\omega t = \pi$, phase voltage V_R is zero, but i_{T1} (or i_R) is not zero because of RL load.
- Therefore, T_1 would continue conducting beyond $\omega t = \pi$.



- When T_1 is on, $V_{T1} = V_R - V_R = 0$ from $\omega t = 75^\circ$ to 195°
 When T_2 is on, $V_{T1} = V_R - V_Y$ from $\omega t = 195^\circ$ to 315° and
 When T_3 is on, $V_{T1} = V_R - V_B$ for $\omega t = 315^\circ$ to 435° and so on.
 If the firing angle is $90^\circ < \alpha < 180^\circ$.
 Let α be 165° ,
- For $\alpha < 90^\circ$, 3- ϕ three pulses converter operates as a line commutated inverter which

is possible only if the load circuit has a dc voltage source of reverse polarity.



Worked out Example

18. A three phase half wave phase controlled rectifier is operated from a 3-phase star connected 400V, 50 Hz supply and the load resistance is $R = 10 \Omega$. If it is required to obtain an average output voltage of 50% of maximum possible output voltage then the converter circuit need to operate at $\alpha =$
- (a) 135° (b) 67.7°
 (c) 60° (d) 30°

Sol: Maximum output voltage $V_{dc \max} = \frac{2\sqrt{3}}{2\pi} V_m$

When load is resistive $\Rightarrow i_o$ is max when

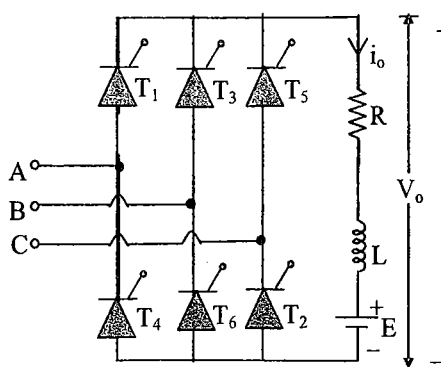
$$V_{dc} = \frac{3V_m}{2\pi} \left(1 + \cos \left(\alpha + \frac{\pi}{6} \right) \right)$$

$$\Rightarrow \frac{V_{dc}}{V_{dc \max}} = 0.5$$

$$= \frac{1}{\sqrt{3}} \left[1 + \cos \left(\alpha + \frac{\pi}{6} \right) \right]$$

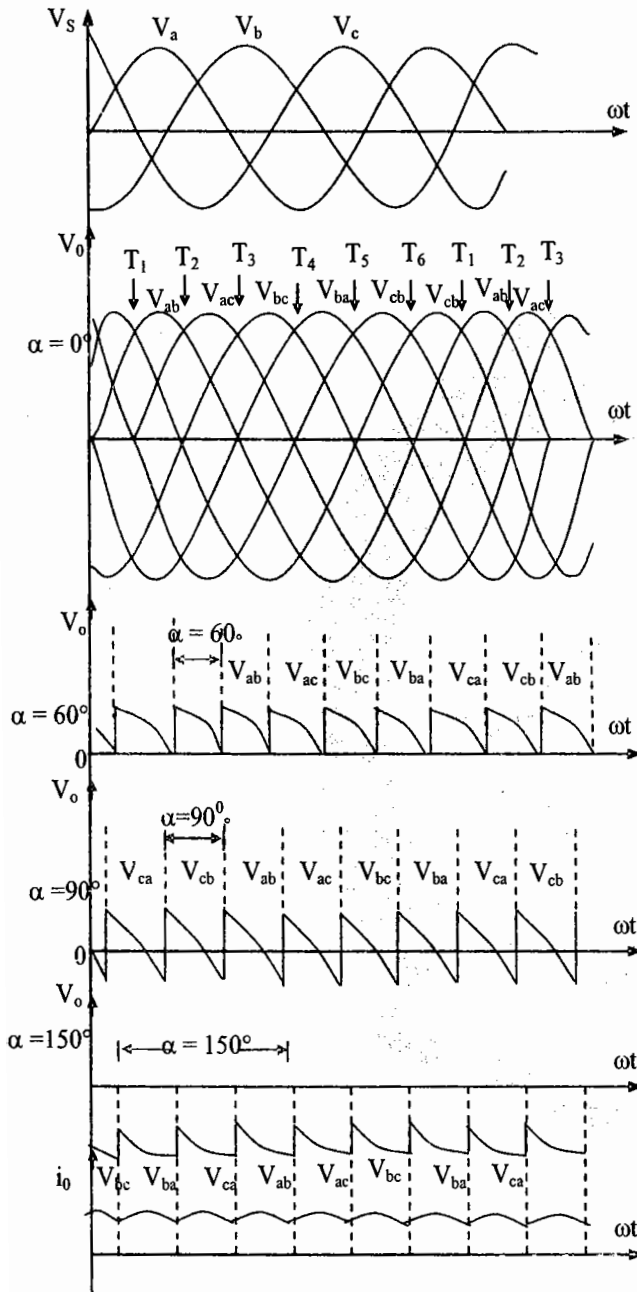
$$\alpha = 67.7^\circ$$

3- ϕ Full Converter (or) 3- ϕ Full wave Phase Controlled Rectifier with RLE Load





- The above circuit works as ac to dc converter for firing angle $0^\circ < \alpha \leq 90^\circ$ and as 3 - ϕ Line commutated inverter for $90^\circ < \alpha \leq 180^\circ$.



- for $\alpha = 0^\circ$
 T_1 is triggered at $\omega t = \frac{\pi}{6}$,
 T_2 at 90° , T_3 at 150° so on.
- for $\alpha = 60^\circ$
 T_1 is triggered at $\omega t = 30^\circ + 60^\circ = 90^\circ$,

T_2 is triggered at $\omega t = 90^\circ + 60^\circ = 150^\circ$, so on,

- For $\alpha = 90^\circ$
 The load voltage is symmetrical about the reference line ωt therefore its average value is zero.
- For $\alpha = 150^\circ$. T_1 is triggered at $\omega t = 210^\circ$, T_2 at 270° .
 The output voltage wave form is reversed in polarity. This means dc source is delivering power ac source, this is called line commutated inverter operation of the 3 - ϕ full converter bridges.
- It should be noted that direction of circuit for both converter and inverter operation remains fixed but the polarity of the output voltage reverse.
- In full wave conversion V_{mL} is taken as maximum voltage.

$$\begin{aligned}
 V_0 &= \frac{1}{T} \int_0^T V_{mL} \sin \omega t \, d\omega t \\
 &= \frac{1}{\pi} \int_{\frac{\pi}{3} + \alpha}^{\frac{2\pi}{3} + \alpha} V_{mL} \sin \omega t \, d\omega t \\
 &= \frac{3 V_{mL}}{\pi} \left[-\cos \omega t \left| \frac{2\pi}{3} + \alpha \right| \right] \\
 &= \frac{3 V_{mL}}{\pi} \left[-\cos \left(\frac{2\pi}{3} + \alpha \right) + \cos \left(\frac{\pi}{3} + \alpha \right) \right] \\
 &= \frac{3 V_{mL}}{\pi} \left[\cos \left(\alpha - \frac{\pi}{3} \right) + \cos \left(\alpha + \frac{\pi}{3} \right) \right] \\
 V_0 &= \frac{3 V_{mL}}{\pi} \cos \alpha
 \end{aligned}$$

Rms value of output voltage V_{orms}

$$\begin{aligned}
 V_{or} &= \sqrt{\frac{3}{\pi} \int_{\frac{\pi}{3} + \alpha}^{\frac{2\pi}{3} + \alpha} V_{mL}^2 \sin^2 \omega t \, d\omega t} \\
 &= \sqrt{\frac{3 V_{mL}^2}{2\pi} \left[\int_{\frac{\pi}{3} + \alpha}^{\frac{2\pi}{3} + \alpha} 1 \, d\omega t - \int_{\frac{\pi}{3} + \alpha}^{\frac{2\pi}{3} + \alpha} \cos 2\omega t \, d\omega t \right]}
 \end{aligned}$$



$$= \sqrt{\frac{3 V_{mL}^2}{2\pi} \left[\left(\frac{\pi}{3} \right) - \frac{1}{2} \left(\sin \left(\frac{4\pi}{3} + 2\alpha \right) - \sin \left(\frac{2\pi}{3} + 2\alpha \right) \right) \right]}$$

$$= \sqrt{\frac{3 V_{mL}^2}{2\pi} \left[\frac{\pi}{3} + \frac{\sqrt{3}}{2} \cos 2\alpha \right]}$$

$$V_{or} = V_{mL} \sqrt{\frac{3}{2\pi} \left[\frac{\pi}{3} + \frac{\sqrt{3}}{2} \cos 2\alpha \right]}$$

Workedout Examples

19. A 3- ϕ full converter feeds power to a resistive load of 10Ω . For a firing angle of 30° the load takes 5kW . Find the magnitude of per phase input supply voltage

(a) 230 V (b) 188 V
(c) 108 V (d) 335.5V

Sol: $V_{or} = \left[\frac{1}{\pi/3} \int_{\frac{\pi}{3} + \alpha}^{\frac{2\pi}{3} + \alpha} V_{mL}^2 \sin^2 \omega t d\omega t \right]^{\frac{1}{2}}$

$$= V_{mL} \sqrt{\frac{3}{2\pi} \left[\frac{\pi}{3} + \frac{\sqrt{3}}{2} \cos 2\alpha \right]}$$

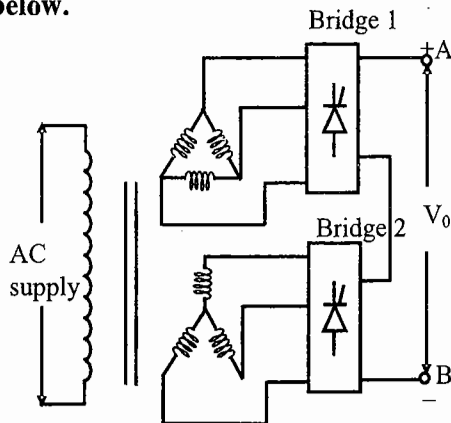
Given that $\frac{V_{or}^2}{R} = 5000 \text{ W}$

$\Rightarrow$ by equating this $V_s = 188.08 \text{ V}$

$\therefore$ per phase supply voltage = 108 V

Common Data for Questions 20 & 21

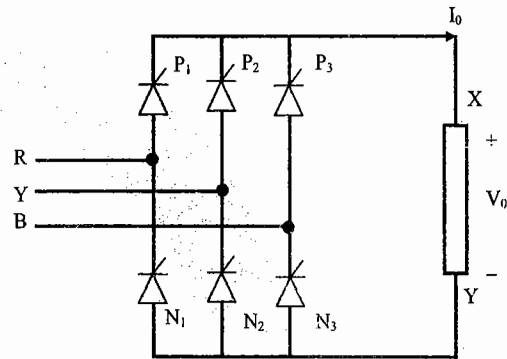
Two 3-phase bridge rectifiers are connected as shown below.



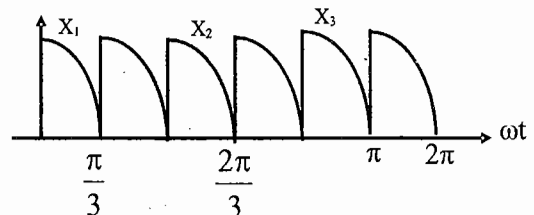
20. The total number of pulses in the output voltage at the terminals A & B is
(a) 6 (b) 12 (c) 3 (d) 24

21. The lowest harmonic in the input current is
(a) 3 (b) 5 (c) 11 (d) 13

22. A 3-phase full converter supplying power to inductive load with ripple free current is shown in fig. All positive group devices are represented with P_1, P_2, P_3 and all negative group devices are represented with N_1, N_2, N_3 as shown below.



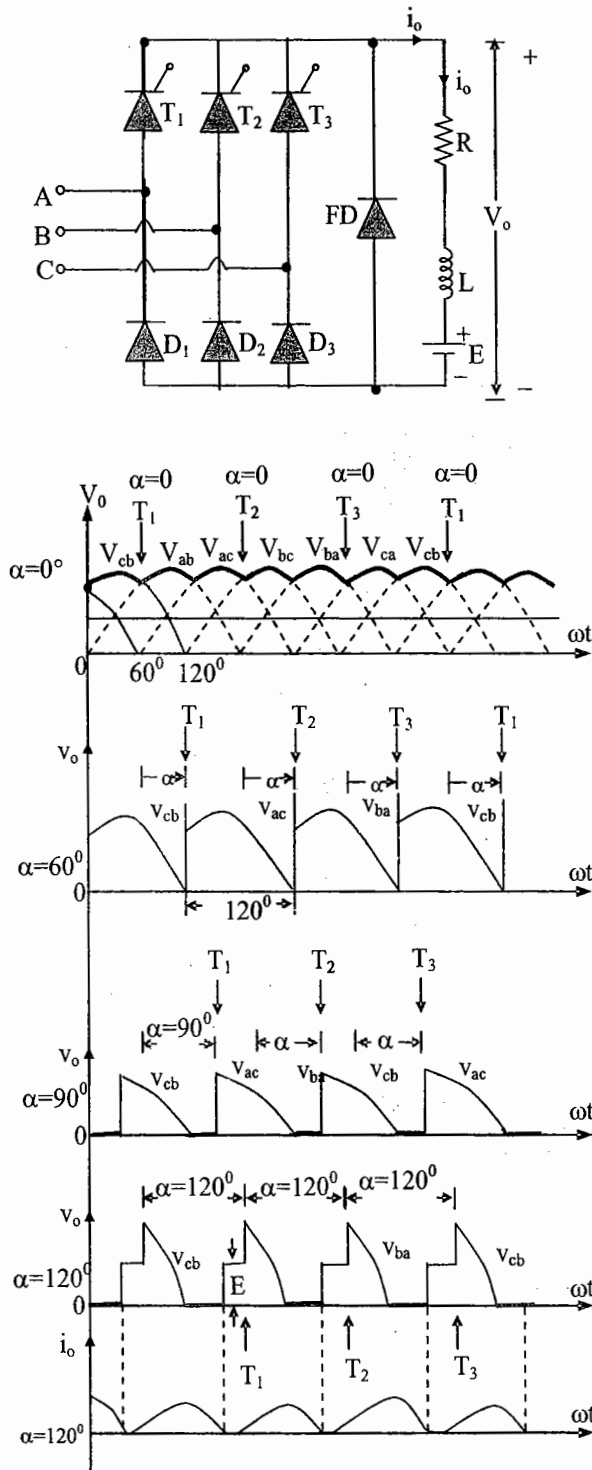
By assuming $V_{YB} = V_{mc} \sin \omega t$ and $\alpha = 60^\circ$ the following load voltage is obtained.



Which of the following statement is true as per the given output voltage waveform

- (a) $X_1 = V_{RB}$, $X_2 = V_{BY}$ & $X_3 = V_{YR}$
(b) $X_1 = V_{RY}$, $X_2 = V_{YB}$ & $X_3 = V_{BY}$
(c) $X_1 = V_{RB}$, $X_2 = V_{YR}$ & $X_3 = V_{BY}$
(d) $X_1 = V_{RY}$, $X_2 = V_{YB}$ & $X_3 = V_{BR}$

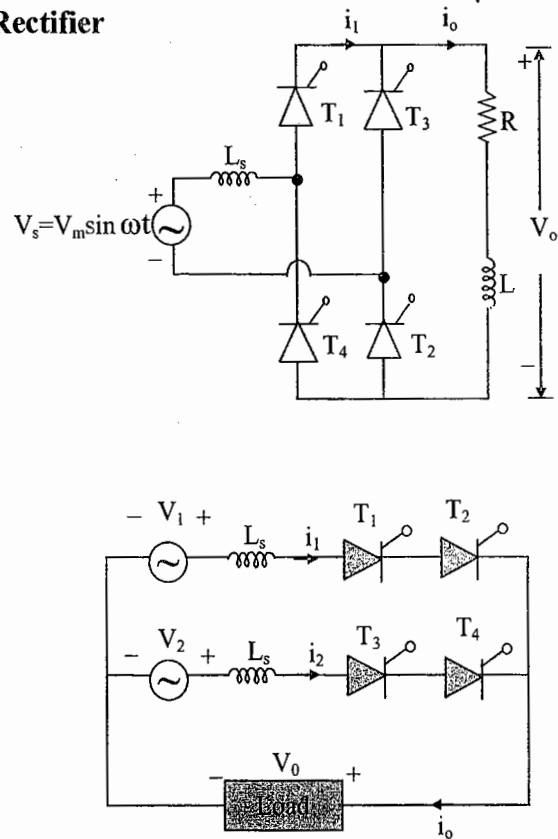
3- ϕ Full Wave Semi Phase Controlled rectifier with RLE Load & Freewheeling Diode:



- An SCR with zero degree firing angle ($\alpha = 0^\circ$), It should be measured from $\omega t = 60^\circ$ for T_1 , from $\omega t = 180^\circ$ for T_2 and from $\omega t = 300^\circ$ to T_3 and so on.

- For $\alpha = 60^\circ$ the thyristor is fired so that current returns through one diode during each 120° conduction period.
- For $\alpha = 90^\circ$ the output voltage V_o is discontinuous for each periodic cycle of 120° , output voltage is equal to line voltage for only 90° and for the remaining 30° , when FD conducts, $V_o = 0$.

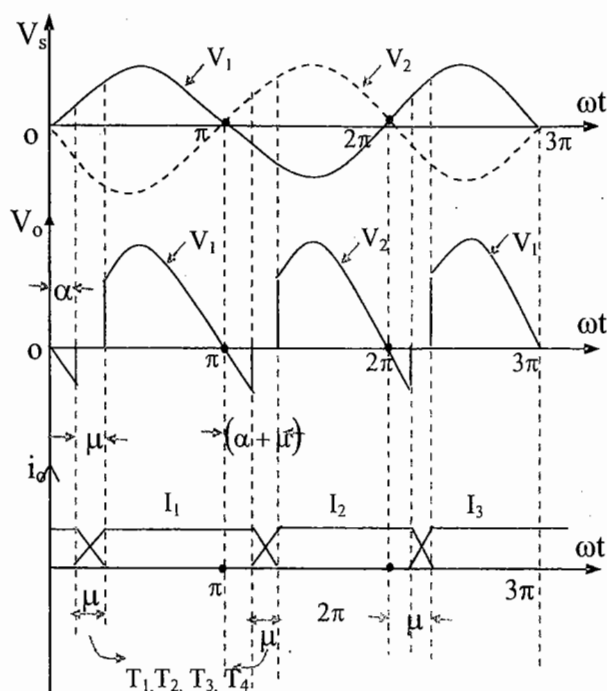
Effect of source inductance in 1 - ϕ full wave Rectifier



- In the full wave rectifier, when conduction shifts from $T_1 T_2$ to $T_3 T_4$, all four thyristors are in conduction which short circuits the source. To limit the short circuit current, L_s is included in the circuit.
- The angular period, during which both the incoming and outgoing SCR's are conducting, is known as commutation angle or overlap angle (μ) in degrees or radians.
- The effect of source inductance is to make the simultaneous conduction of incoming & outgoing SCR's at the time of transition. In this period output voltage is equal to zero.



- The net effect of source inductance is to decrease the average value of output voltage.



By applying KVL,

$$V_1 - L_s \frac{di_1}{dt} = V_2 - L_s \frac{di_2}{dt}$$

$$V_1 - V_2 = L_s \left(\frac{di_1}{dt} - \frac{di_2}{dt} \right)$$

$$\frac{di_1}{dt} - \frac{di_2}{dt} = \frac{V_1 - V_2}{L_s} = \frac{2V_m \sin \omega t}{L_s}$$

$$\frac{di_1}{dt} - \frac{di_2}{dt} = \frac{2V_m \sin \omega t}{L}$$

As the load current is assumed constant through out, $i_1 + i_2 = I_o$

Differentiating this with respect to t , we get

$$\frac{di_1}{dt} - \frac{di_2}{dt} = \frac{dI_o}{dt} = 0$$

Adding above two equations,

$$\frac{di_1}{dt} = \frac{V_m}{L_s} \sin \omega t$$

$$\int_0^{I_o} di_1 = \int_{\alpha/\omega}^{\frac{\alpha+\mu}{\omega}} \frac{V_m}{L_s} \sin \omega t \cdot dt$$

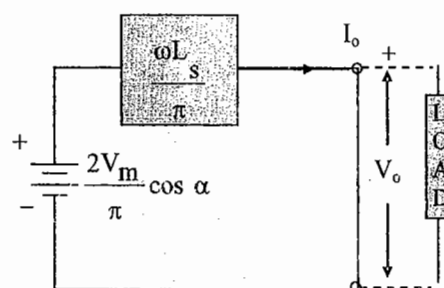
$$I_o = \frac{V_m}{\omega L_s} [\cos \alpha - \cos(\alpha + \mu)]$$

$$V_o = \frac{1}{\pi} \int_{\alpha}^{\alpha+\mu} V_m \sin \omega t \cdot d(\omega t)$$

$$V_o = \frac{V_m}{\pi} [\cos \alpha + \cos(\alpha + \mu)]$$

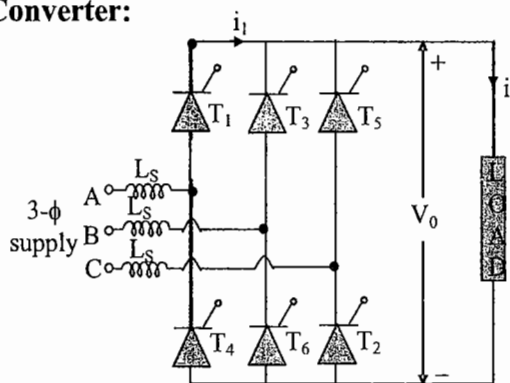
$$\cos(\alpha + \mu) = \cos \alpha - \frac{L_s \omega}{V_m} I_o$$

$$\therefore V_o = \frac{2V_m}{\pi} \cos \alpha - \frac{\omega L_s}{\pi} I_o$$



- The inductance of source results in an lesser value of voltage.
- Inductive voltage regulation = $\left[\frac{1 - \cos \mu}{2} \right]$

Effect of source Inductance 3 - ϕ Full wave Converter:



$$V_o = \frac{3\sqrt{6}}{\pi} V_{ph} \cos \alpha - \frac{3\omega L_s}{\pi} I_o$$

$$\cos(\alpha + \mu) = \cos \alpha - \frac{2\omega L_s}{V_{mL}} I_o$$

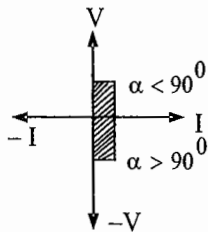


- In 3 - ϕ full converter, voltage regulation due to source inductance

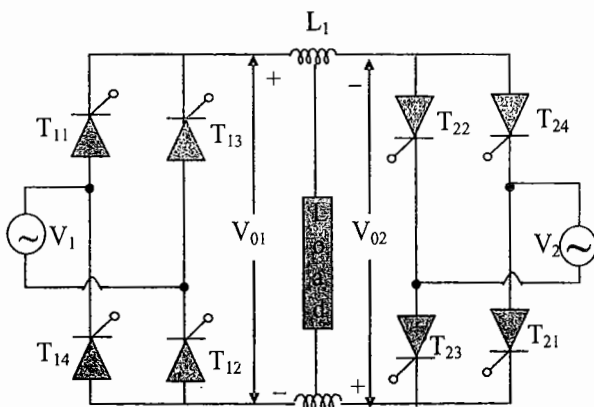
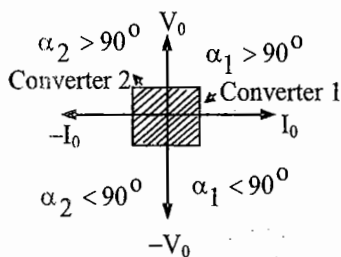
$$= \frac{2\pi f \cdot L_s \cdot I_o}{V_{mL} \cdot \cos \alpha}$$

Dual Converter:

- The 1 - ϕ phase controlled converter operates only in two quadrants i.e, I & IV.



- Here in this voltage is reversing but the current cannot reverse.
- There are two converters connected back to back antiparallely hence the name dual converter.
- using a dual converter, all four quadrant operation is possible converter 2



$$V_{o1} = \frac{2V_m}{\pi} \cos \alpha_1$$

where α_1 is the firing angle of converter 1.

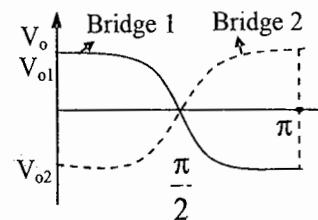
$$V_{o2} = \frac{2V_m}{\pi} \cos \alpha_2$$

Where α_2 is the firing angle of converter 2.

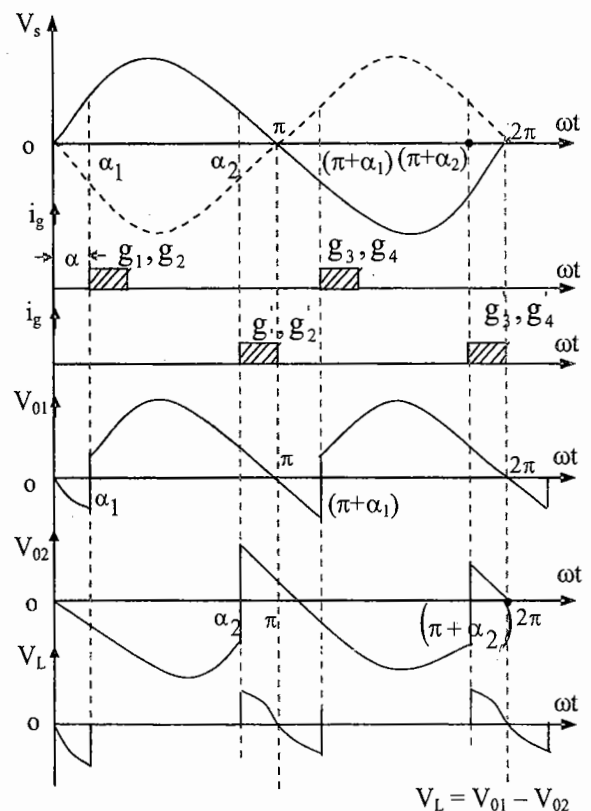
by KVL : $V_{o1} = -V_{o2}$

$$V_{o1} + V_{o2} = 0$$

$$\alpha_2 = \pi - \alpha_1$$



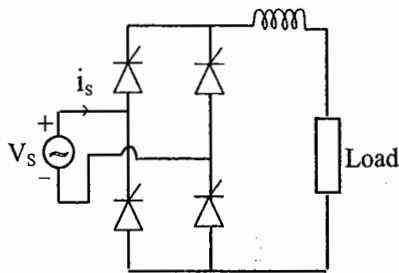
Output wave forms of a dual converter:





Worked out Examples

23. A fully controlled converter bridge feeds a highly inductive load with ripple free load current. The input supply (V_s) to the bridge is a sinusoidal source. Triggering angle of the bridge converter is $\alpha = 30^\circ$. The input power factor of the bridge is _____.



Sol: $\alpha = 30^\circ$

Input power factor of the bridge = distortion factor $\times$ displacement factor

$$\Rightarrow \text{IPF} = \frac{2\sqrt{2}}{\pi} \times \cos \alpha = 0.9 \times \cos 30^\circ = 0.7794$$

lag

24. A 3- ϕ Fully controlled rectifier is operated from a 3- ϕ star connected 400 V, 50 Hz AC supply and the load resistance is $R = 10\Omega$. A large inductance is connected in series with the load to maintain ripple free load current. If it is required to obtain an average output voltage of 86.66% of the maximum possible output voltage. Find the firing angle

- (a) 30° (b) 60°
(c) 45° (d) 90°

Sol: $V_{0 \max} = \frac{3V_{ml}}{\pi}$

$$V_0 = \frac{3V_{ml}}{\pi} \cos \alpha \Rightarrow \alpha = 30^\circ$$